

EXERCISE 4

SELECT order-id, customer-id, order-date,
 DAYNAME (order-date) AS day-name
 FROM orders;

order-id	customer-id	order-date	day-name
1001	101	2026-05-01	Friday
1002	102	2026-05-02	Saturday
1003	103	2026-05-03	Sunday
1004	104	2026-05-04	Monday
1005	105	2026-05-05	Tuesday

SELECT customer-id, customer-name, ~~signup-date~~,
 MONTHNAME (signup-date) AS signup-month-name
 FROM customer-signups;

customer-id	customer-name	signup-date	signup-month-name
201	John	2026-01-15	January
202	Mary	2026-02-20	February
203	Peter	2026-03-05	March
204	Sarah	2026-04-18	April
205	Thabo	2026-05-30	May

③ ~~sale-id~~ SELECT sale-id, product-name, sale-date,
MONTH(sale-date) AS sale-month
FROM sales;

sale-id	product-name	sale-date	sale-month
S001	Laptop	2026-01-10	1
S002	Mouse	2026-02-15	2
S003	Keyboard	2026-03-20	3
S004	Monitor	2026-04-25	4
S005	Desk	2026-05-30	5

④ SELECT transaction-id, customer-id, transaction-date,
YEAR(transaction-date) AS transaction-year
FROM transactions;

transaction-id	customer-id	transaction-date	transaction-year
T001	101	2024-12-15	2024
T002	102	2025-01-20	2025
T003	103	2025-06-10	2025
T004	104	2026-02-05	2026
T005	105	2026-05-25	2026

SELECT delivery-id, customer-id, delivery-date,
 DAY (delivery-date) AS day-of-month
 FROM deliveries;

delivery-id	customer-id	delivery-date	day-of-month
D001	101	2026-05-01	1
D002	102	2026-05-08	8
D003	103	2026-05-15	15
D004	104	2026-05-22	22
D005	105	2026-05-29	29

SELECT employee-id, employee-name, department,
 CURRENT-DATE() AS today-date
 FROM employees;

employee-id	employee-name	department	today-date
301	Mandi	Sales	2026-07-19
302	Brian	IT	2026-07-19
303	Lerato	Finance	2026-07-19
304	Siphiso	HR	2026-07-19
305	Aisha	Marketing	2026-07-19

⑦ SELECT order-id, customer-id, order-date-text,
TO-DATE (order-date-text, 'YYYY-MM-DD') AS order-date
FROM online-orders;

order-id	customer-id	order-date-text	order-date
4001	101	2026-01-13	2026-01-13
4002	102	2026-02-20	2026-02-20
4003	103	2026-03-25	2026-03-25
4004	104	2026-04-10	2026-04-10
4005	105	2026-05-05	2026-05-05

⑧ SELECT payment-id, customer-id, payment-date,
TO-CHAR (payment-date, 'YYYY-MM-DD') AS
formatted-payment-date
FROM payment-sales;

payment-id	customer-id	payment-date	Formatted-payment-date
P001	101	2026-01-05	2026-01-05
P002	102	2026-02-10	2026-02-10
P003	103	2026-03-15	2026-03-15
P004	104	2026-04-20	2026-04-20
P005	105	2026-05-25	2026-05-25

9 SELECT customer-id, customer-name, last-purchase-date
 DATEDIFF(CURRENT-DATE(), last-purchase-date) AS
 day-since-last-purchase
 FROM customer-purchase;

customer-id	customer-name	last-purchase-date	day-since-last-purchase
501	John	2026-05-01	79
502	Mary	2026-05-10	70
503	Peter	2026-05-15	65
504	Sarah	2026-05-20	60
505	Thabo	2026-05-25	55

10 SELECT order-id, customer-id, order-date,
 DATEADD(DAY, 7, order-date) AS expected-
 delivery-date
 FROM shipping-orders;

order-id	customer-id	order-date	expected-delivery-date
6001	101	2026-05-01	2026-05-08
6002	102	2026-05-03	2026-05-10
6003	103	2026-05-05	2026-05-12
6004	104	2026-05-07	2026-05-14
6005	105	2026-05-09	2026-05-16

⑪ SELECT booking-id, customer-id, booking-date,
 YEAR(booking-date) AS booking-year,
 MONTH(booking-date) AS booking-month,
 DAY(booking-date) AS booking-day
 FROM bookings;

booking-id	customer-id	booking-date	booking-year	booking-month	booking-day
B001	101	2026-01-12	2026	1	12
B002	102	2026-02-18	2026	2	18
B003	103	2026-03-22	2026	3	22
B004	104	2026-04-09	2026	4	9
B005	105	2026-05-27	2026	5	27

⑫ SELECT order-id, customer-id, order-date,
 YEAR(order-date) AS order-year, amount
 FROM party-orders
 WHERE (order-date) = 2026;

order-id	customer-id	order-date	order-year	amount
7004	104	2026-02-05	2026	1500
7005	105	2026-05-25	2026	2000

⑬ SELECT order-id, customer-id, order-date,
 MONTH(order-date) AS order-month, amount
 FROM monthly-orders
 WHERE MONTH(order-date) = 3;

order-id	customer-id	order-date	order-month	amount
8003	103	2026-03-10	3	800
8004	104	2026-03-25	3	1500

⑭ SELECT subscription-id, customer-id, start-date,
 LAST-DAY(start-date) AS month-end-date
 FROM subscriptions;

subscription-id	customer-id	start-date	month-end-date
SUB001	101	2026-01-10	2026-01-31
SUB002	102	2026-02-15	2026-02-28
SUB003	103	2026-03-20	2026-03-31
SUB004	104	2026-04-25	2026-03-30
SUB005	105	2026-05-30	2026-05-31

⑮ SELECT send-id, customer-id, send-date,
 DATE_TRUNC('month', send-date) AS month-start-date
 FROM campaign-send;

send_id	customer_id	send_date	month_start_date
C001	101	2026-01-12	2026-01-01
C002	102	2026-02-18	2026-02-01
C003	103	2026-03-22	2026-03-01
C004	104	2026-04-09	2026-04-01
C005	105	2026-05-27	2026-05-01

(16) SELECT invoice_id, customer_id, invoice_date,
TO_CHAR(invoice_date, 'Month YYYY') AS invoice_month
FROM invoice_dates;

invoice_id	customer_id	invoice_date	invoice_month_year
INV001	101	2026-01-05	January 2026
INV002	102	2026-02-10	February 2026
INV003	103	2026-03-15	March 2026
INV004	104	2026-04-20	April 2026
INV005	105	2026-05-25	May 2026

(17) SELECT customer_id, customer_name, date_of_birth,
FLOOR(~~date~~DATE - 177 * (CURRENT DATE), date_of_birth)
365, 25) AS customer_age
FROM customer_birthdays;

customer-id	customer-name	date of birth	customer-age
9001	John	1995-05-10	28
9002	Mary	1990-08-20	35
9003	Peter	2002-03-15	24
9004	Sarah	1985-12-01	40
9005	Thabo	2000-07-30	25

15) SELECT order-id, customer-id, order-date,
 DATENAME(order-date) AS day-name,
 CASE
 WHEN DATENAME(order-date) IN ('Saturday', 'Sun-
 day') THEN 'Weekend'
 ELSE ~~Weekend~~ 'Weekday'
 END AS day-type
 FROM weekend-orders;

order-id	customer-id	order-date	day-name	day-type
9001	101	2026-05-01	Friday	Weekday
9002	102	2026-05-02	Saturday	Weekend
9003	103	2026-05-03	Sunday	Weekend
9004	104	2026-05-04	Monday	Weekday
9005	105	2026-05-05	Tuesday	Weekday

(19) SELECT transaction-id, customer-id, transaction-date,
 QUARTER(transaction-date) AS transaction-quarter,
 amount;

FROM quarterly-transactions;

transaction-id	customer-id	transaction-date	transaction-quarter	Amount
R001	101	2026-01-13	1	500
R002	102	2026-03-20	1	1200
R003	103	2026-04-10	2	800
R004	104	2026-07-05	3	1500
R005	105	2026-10-27	4	2000

(20) SELECT order-id, customer-id, order-date,
 DATEDIFF(CURRENT_DATE(), order-date) AS
 days-since-order, amount
 FROM recent-orders
 WHERE DATEDIFF(CURRENT_DATE(), order-date)

730;

Order-id	customer-id	order-date	days-since-first-order	amount
R001	101	2026-04-01	109	500
R002	102	2026-04-15	95	1200
R003	103	2026-05-01	79	800
R004	104	2026-05-10	70	1500

BONUS

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SELECT customer_id, customer_name, last_purchase_date,  
DATEIF7(current_date(), last_purchase_date)  
AS days-since-last-purchase,  
CASE  
WHEN DATEIF7(current_date(), last_purchase_date) <= 30  
THEN 'Active Customer'  
WHEN DATEIF7(current_date(), last_purchase_date) <= 90  
THEN 'At Risk Customer'  
END AS ELSE 'Inactive Customer'  
END AS customer_status  
FROM customer_recency;
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customer_id	customer_name	last_purchase_date	day-since-last-purchase	customer_status
1001	John	2026-05-25	55	At Risk Customer
1002	Mary	2026-05-10	70	At Risk Customer
1003	Peter	2026-04-01	109	Inactive Customer
1004	Sarah	2026-02-15	154	Inactive Customer
1005	Thabo	2026-12-20	211	Inactive Customer